

Commissioning a high-repetition rate, 1 J Nd:YAG laser upgrade for the HSX Thomson scattering system

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Thomson scattering (TS) diagnostics are ubiquitous among high-temperature plasma experiments because they provide electron temperature and density measurements with excellent spatial resolution and well-characterized uncertainties. TS signal levels are intrinsically low (scattered power fractions on the order of 10^{-14}), so high-energy nanosecond laser pulses and photon-statistics-limited inference are required. In many TS systems, the laser repetition rate is a primary limiter on temporal coverage within a discharge and on the ability to reduce random uncertainties through averaging.

We report on the installation and commissioning of a major TS upgrade at the HSX stellarator, centered on a new Nd:YAG laser capable of burst operation of 80 consecutive 1 J, 10 ns pulses at 400 Hz. A 2 kHz mode supports up to 10 pulses per burst, limited by thermal loading. The upgrade includes integrating the laser with HSX beam-delivery optics and shot-to-shot pulse-energy monitoring to verify stability. To accommodate the increased data rate, a high-speed digitizer (3.2 GSa/s) and low-noise preamplifiers, providing an ~ 8.6 dB improvement in signal-to-noise ratio over the legacy digitizer chain [1]. In the near term, increasing the time resolution of the TS system will enable the study of plasma breakdown effects, the island-healing effect [2], impurity injection behavior, and time-resolved density profile adjustments.

In parallel, we use filter polychromators with multiple spectral passbands to improve spectral utilization and reduce inferred TS uncertainties without increasing detector count [3]. For HSX, this enables a compact configuration where a system with only three multi-passband filters has equivalent utility (temperature sensitivity and density precision over the HSX T_e range) to a standard five-channel single-passband system, while maintaining or improving uncertainty relative to a traditional single-passband filter set. The higher shot rate increases statistical power and improves temporal resolution, while the multi-passband filters increase spectral utilization per shot. Together, these changes reduce uncertainty associated with a detector set and discharge time.

Commissioning results and first TS measurements from the upgraded system will be presented, with emphasis on the laser installation, high-repetition-rate operation, and resulting improvements in temporal resolution compared with the prior configuration.

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[1] W. R. Goodman et al, Rev. Sci. Instrum. 93, 093518 (2022)

[2] C. C. Hegna, Phys. Plasmas 19, 056101 (2012)

[3] W. R. Goodman, B. Geiger, Rev. Sci. Instrum. 95, 093507 (2024)

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